Remanent Magnetization of Some Upper Cretaceous Granitic Plutons in the Sierra Nevada, California

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Abstract. Extensive sampling of the granitic plutons in Yosemite National Park, California, led to the retention of nine sites distributed in three plutons (Gateway, Sentinal, Half Dome) ranging in age (K-Ar) from about 89 to 79 million years (early Upper Cretaceous). Many specimens are unstable; they carry, in addition to their thermal remanence, a 'random' component of unknown origin. Mean directions for large numbers of specimens seem, however, to be significant. The mean pole for that time falls at 73.5°N, 174°E, in agreement with previous results. Virtual poles for the several sites are scattered with an angular deflection factor $\delta = 10.2°$; they appear to show a clockwise (east to west) rotation of the geomagnetic pole about the mean rotational pole with a period of 10^5 to 10^6 years. There are some indications that the secular variation is not entirely smoothed out in a time of the order of 10^5 years.

INTRODUCTION

The great granite batholith of the Sierra Nevada, California, consists of a number of smaller plutons successively emplaced. Potassium-argon ages establish an absolute chronology which agrees in general with the order of intrusion deduced from field observations. All dates correspond to the early Upper Cretaceous, a time for which paleomagnetic data are still very few. It was hoped that a study of the paleomagnetism of these plutons might provide a reliable geomagnetic pole for that time, and that it might be possible to use the radioactive ages to obtain some idea of the behavior of the earth's field at known intervals of time and of the time constants of the secular variation. The region of Yosemite Valley was chosen because it has been more extensively studied geologically than the rest of the Sierra Nevada.

GEOLOGIC SETTING

Granitic rocks in the region of Yosemite Valley were first described and mapped in detail by *Calkins* [1930], who noted the complexity of the immediate area of the valley as contrasted with the surrounding area and emphasized the composite nature of the Sierra Nevada batholith. Calkins' map was later extended by *Rose* [1957], who described the area around May Lake in detail. It was found by these workers that the granitic rocks could be mapped as discrete homogeneous bodies or plutons, distinguished from one another on the basis of mineralogy and texture. They established the time sequence of intrusion of the plutons by the following criteria: (1) The older rock is penetrated by dikes or apophyses of the younger; (2) fragments of the older rock are enclosed in the younger; (3) foliation defined by alignment of tabular minerals (usually plagioclase and biotite) in the older pluton is truncated by the margins of the younger; and (4) similar foliation in the younger pluton is generally parallel to its margins. These structural relations, the first two of which can be seen clearly in Figure 1, are commonly considered to be convincing field evidence of the intrusive magmatic origin of the granitic rocks (as opposed to a possible metasomatic origin).

The granitic rocks sampled for this study range in composition from diorite to quartz monzonite and are referred to, collectively and loosely, as 'granites.' Petrographic descriptions of them have been given by *Calkins* [1930] and *Rose* [1957]; they contain zoned plagioclase, microcline (often perthitic), and quartz in varying proportions, with biotite and subordinate hornblende as the essential mafic minerals. The magnetite content is about 1 to 2 per cent by weight, and appreciable ilmenite does not occur separately from the magnetite in any of the rocks sampled here [*Pabst*, 1938]. Abundant accessory sphene is present, often surrounding magnetite grains; it probably accounts for most or all of the TiO_2 content which ranges from trace

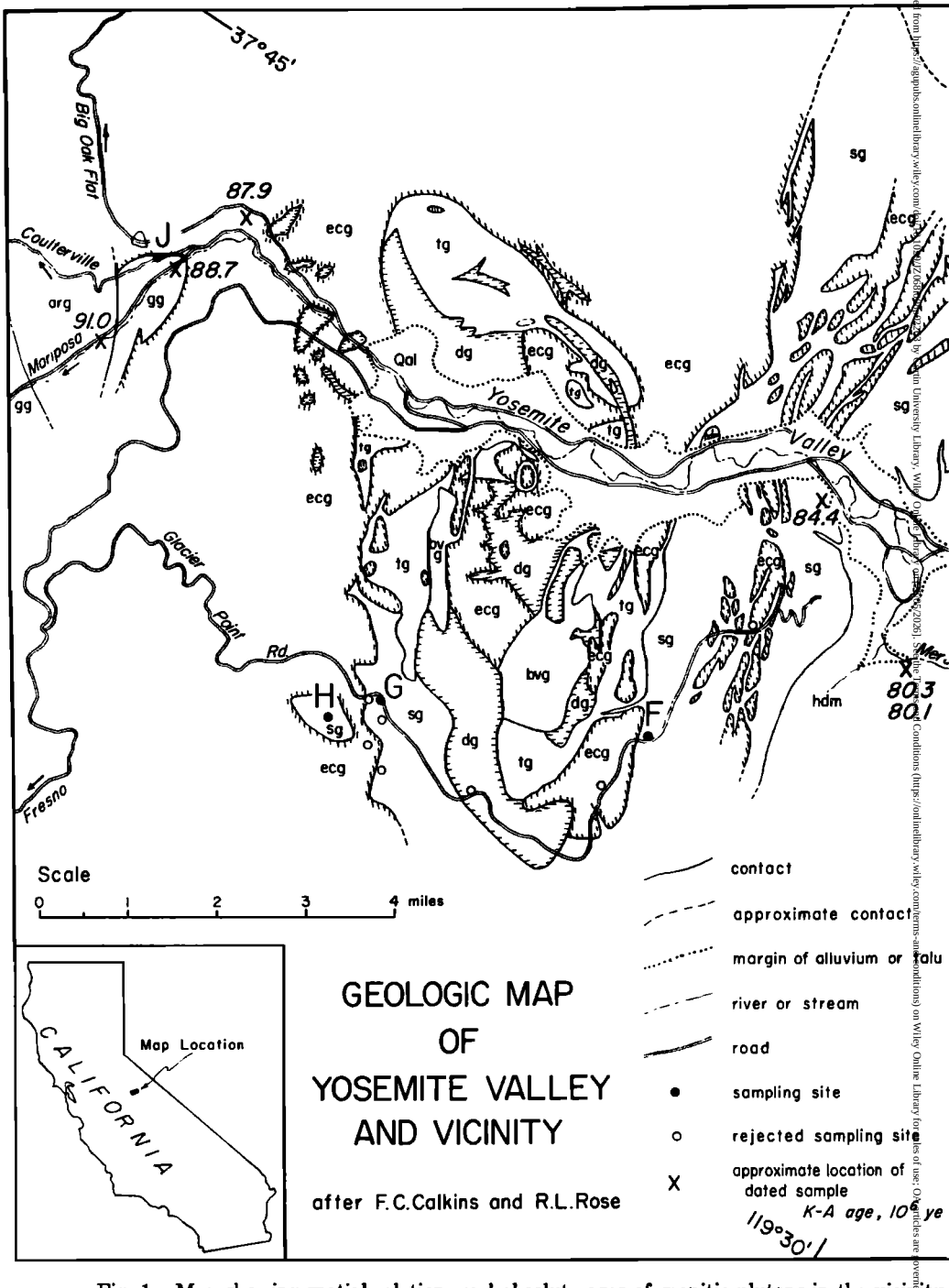
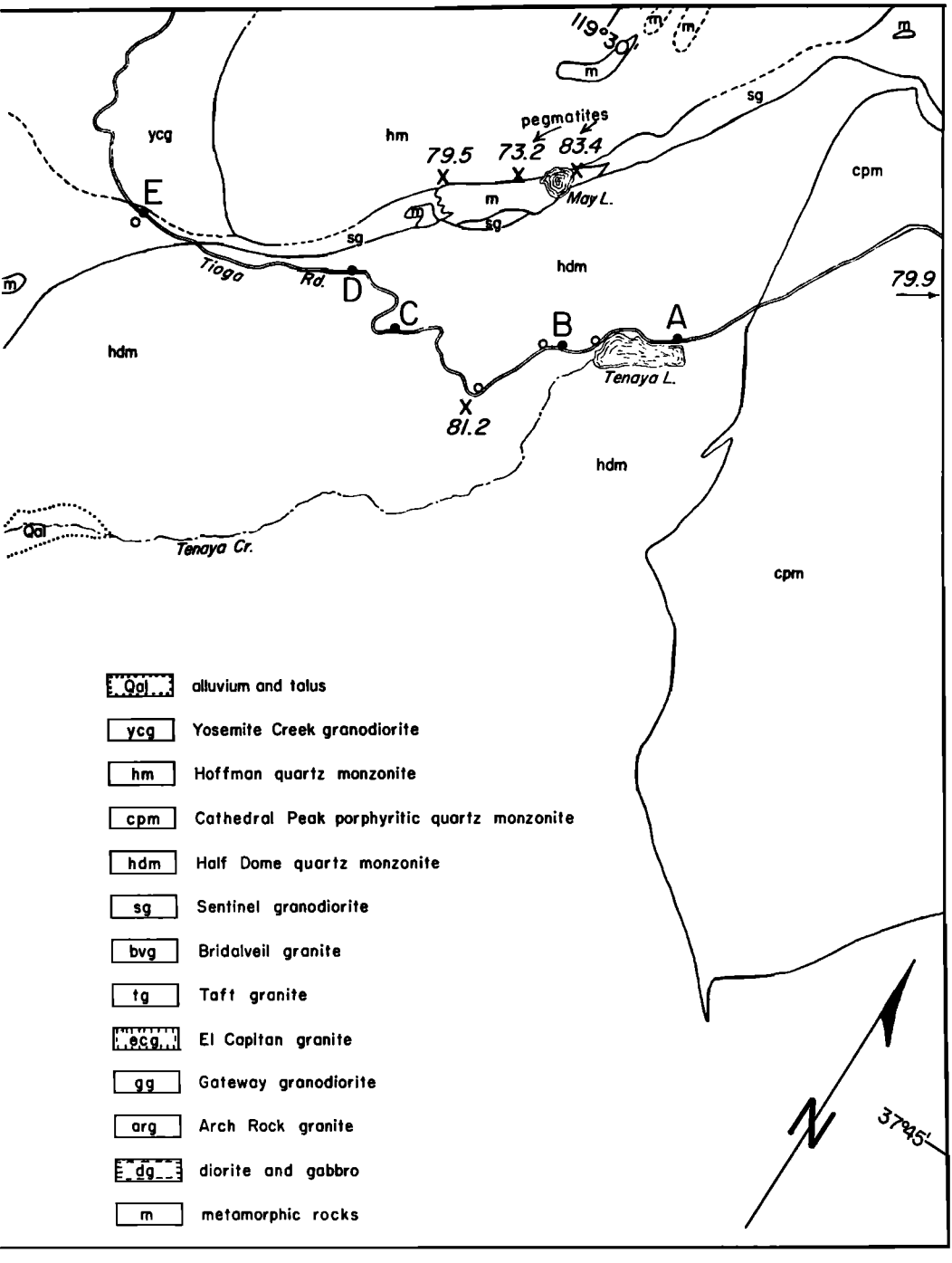


Fig. 1. Map showing spatial relations and absolute ages of granitic plutons in the vicinity

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osemite Valley, with sampling localities. K-Ar ages recalculated from *Evernden et al.* [1957].

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TABLE 1. Cell Dimensions of Magnetites Separated from Granitic Rocks

Cubic Cell Edge, angstroms	Rock Type	Name of Pluton
8.394 ± .004	Autolith	Half Dome
8.39 ± .01	Autolith	Half Dome
8.388 ± .006	Quartz monzonite	Half Dome
8.395 ± .005	Autolith	Sentinel
8.393 ± .009	Granodiorite	Sentinel
8.397 ± .003	Autolith	Sentinel
8.396 ± .005	Granodiorite	Sentinel
8.39 ± .01	Granodiorite	Gateway
8.396 ± .0005	Mineral from Bisperg, Sweden*	
8.394 ± .0005	Artificial material*	

* *Nagata* [1961].

to 0.8 per cent by weight in analyses of various Sierran granites reported by *Pabst* [1928]. X-ray diffraction patterns were obtained from eight ferromagnetic concentrates; they showed only peaks corresponding to magnetite and occasional minor amounts of the major minerals. No trace of any rhombohedral oxide was found. Cell dimensions of the magnetites are given in Table 1. No significant differences occur among them, nor between them and those given by other workers and selected by *Nagata* [1961, p. 80] for natural and synthetic pure magnetites. (The large uncertainties given in Table 1 are due (1) to broadening of peaks owing to strains produced in the magnetite during grinding of the rock (the concentrate was not annealed) and (2) to the use of a diffractometer rather than a powder camera. The measured *d* spacings were systematically corrected by means of reference diffractograms run at the same time on two standard magnetites, whose cell dimensions had been accurately determined with a back-reflection focusing powder camera.)

Distinct foliation is evident in the Sentinel pluton in the area from sample site E to the northeast corner of the map. It strikes parallel to the contacts and is essentially vertical; it is defined by preferred orientation of tabular plagioclase and biotite [*Rose*, 1957]. Foliation is much less evident in the Sentinel pluton at sites F, G, and H, and is absent in the other rocks sampled. Melanocratic schlieren occur in many of these rocks, particularly adjacent to contacts, but no samples were taken from them. Cataclasis or other evidence of post-consolidational deformation (except jointing) has not been observed in these rocks, with the exception of the Hoffman quartz monzonite [*Curtis et al.*, 1958], which was not sampled.

Dark-colored inclusions ranging in size from a few centimeters to a meter or more are widely distributed in the Cretaceous granitic rocks throughout the Sierra Nevada. They are absent only in the youngest intrusions (Cathedral Peak porphyritic quartz monzonite and other porphyritic rocks exposed on the crest of the range and on the eastern scarp) and in the diorites and gabbros. The inclusions have been described in detail by *Pabst* [1928], who called them *autoliths*, solely to distinguish them from obvious xenoliths and with no further genetic implications; his usage will be followed here. The following description is taken almost entirely from his paper.

The autoliths are sharply bounded, but are distinguished from the enclosing rock only in having a considerably higher color index and a much smaller grain size. Their texture is hypidiomorphic granular ('granitic'), as is that of the granitic rocks, in marked contrast to the allotriomorphic granular ('hornfelsic') texture of the xenoliths. No gradational types occur. The autoliths are composed of identically the same minerals as the enclosing rocks; the zoning in the plagioclase is the same in both, whereas the xenoliths contain unzoned andesine. Some autoliths are porphyritic, with phenocrysts of zoned plagioclase. Foliation in the autoliths parallels that of the enclosing pluton, which is in turn parallel to the contact with older rock; foliation in xenoliths is generally discordant to that of the surrounding rock. The autoliths are mostly ellipsoidal in outline, and those near the margins of a pluton are more flattened than those near the center. They are often grouped in swarms or bands, but their distribution is unrelated to the shape of the enclosing pluton or, unlike the far less abundant xenoliths, to their proximity to its margin. The physical relation of the autoliths to the melanocratic schlieren is obscure and indicates no common genesis. *Pabst* [1928] concluded that the autoliths were plastically deformed during emplacement of the enclosing granitic rocks, that their distribution appears related to magmatic movements, and that there

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TABLE 2. Possible Chronology of Sampling Sites

Site	Pluton	Age, m.y.
J	Gateway	89
H, G, F, E(?)	Sentinel	86–82
D(?), C, B, A	Half Dome	81–79

is abundant evidence for their not having been derived from any of the presently exposed wall rocks surrounding the batholith. Without speculating on their genesis, we can say that the autoliths have had a cooling history very nearly the same as that of the surrounding granitic rocks, differing in its early (preconsolidation) stages only sufficiently for them to have preserved their chemical composition, grain size, and sharp outlines.

Age Relationships

Potassium-argon ages have been determined on biotites separated from some of these rocks [*Evernden et al.*, 1957]; the values shown in Figure 1 have been recalculated using improved values of the disintegration constants ($\lambda_\epsilon = 0.585 \times 10^{-10}$ yr^{-1} and $\lambda_\beta = 4.72 \times 10^{-10}$ yr^{-1}). In all cases, the sequence of radioactive ages agrees with the sequence of intrusion as mapped and shown in the legend of Figure 1. It is seen that the interval of time between adjacent intrusions varies from 1 to 4 $\times$ 10^6 years and that therefore the time required for solidification (i.e., for the 700°C isotherm to move downward a few kilometers) is of the order of 10^6 years or less [*Curtis et al.*, 1958].

The K-Ar ages refer to the time elapsed since the temperature at the sampling site dropped to a level such that the rate of escape of Ar from biotite became negligible. The rate of diffusion of Ar in phlogopite has been studied by *Evernden et al.* [1960]. They find that a phlogopite grain 1 mm thick would lose 40 per cent of its Ar if held at 320°C for 10^5 years, 4 per cent if held at 225°C for the same time, and 0.4 per cent at 150°C. It follows that the ages shown must refer to the time when the temperature dropped to 200°C or less.

Most of the thermoremanent magnetization (TRM), on the other hand, was probably acquired in the approximate range 580° to 400°; the 'magnetic age' must therefore be older than the K-Ar ages. At none of the collecting sites is it possible to determine exactly the geometric relationship of the pluton to its wall rock and to the surface at the time of intrusion; the thickness of the sedimentary cover, for instance, can only be guessed.[1] Thus one cannot evaluate exactly either the time of magnetization or the time during which TRM was acquired. However, rates of cooling for batholithic bodies as calculated by *Larsen* [1945] may serve as an indication, if one chooses to ignore outward transfer of heat by all processes other than conduction. Imagine, for instance, a batholith intruded at 800°C under a sedimentary cover 1 km thick. The temperature at a point at the top of the batholith would have dropped to 300°C about 100,000 years after intrusion; it could have spent some 20,000 years between 600° and 400°C. A point 2.2 km below the surface (i.e., 1.2 km below the top of the pluton) would cool to 600° in 80,000 years, and to 400° in 300,000 years. A specimen collected 3.8 km below the surface would have remained between 600° and 400° for about 500,000 years. Most of our specimens may have spent from 50,000 to 500,000 years in the range 600° to 400°C, and their magnetic age may exceed their K-Ar age by a few hundred thousand years. The chronological sequence of our sites cannot be determined exactly, but presumably specimens from a higher topographic level are magnetically older than specimens from a lower level, and samples closer to the lateral boundary of a pluton are presumably older than those collected nearer to its center. It should be noted that the range of ages indicated in Figure 1 within the Half Dome quartz monzonite is as great as the difference between the ages of the El Capitan and Gateway plutons, and also that reheating of an older rock at the contact of a younger intrusion may lower its K-Ar age. Thus the age at site E might be that of the Yosemite Creek granodiorite rather than that of the Sentinel body. Even so, reheating that is sufficient to alter a K-Ar age might not be sufficient to change the thermal remanence appreciably, as most of the latter is acquired at a relatively high temperature. There is thus no reason to believe that the magnetic chronology should be much

[1] Guesses range between 3 and 10 km. Our collecting sites lie between 1.2 and 2.8 km below the highest exposure of granite in the area.

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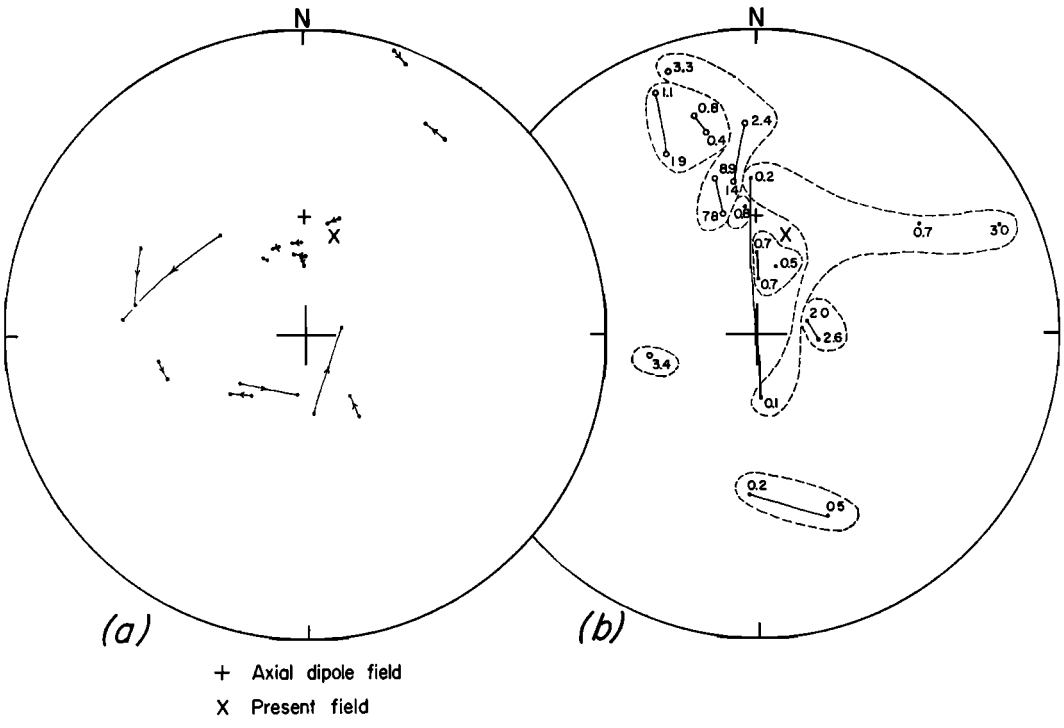


Fig. 2. Equal-area projections of north-seeking directions from two rejected sites. (*a*) Lines connect directions of single specimens before and after storage for 1 month in laboratory. All points in lower hemisphere. (*b*) Lines connect directions of specimens cut from one core. Dashed lines enclose single groups of cores at sampling site. Numbers denote Q, the ratio of natural remanence to the induced magnetization in the earth's field. Circles in upper hemisphere, dots in lower.

different from that based on the apparent order of intrusion. A possible chronological sequence for our sites is given in Table 2. In this chronology, the age at site J (Gateway) is essentially that of the nearby K-Ar sample (88.7 m.y.). Three plutons (El Capitan, Taft, and Bridalveil) were intruded after Gateway and before Sentinel; therefore, Sentinel cannot be much older than 86 or 87 m.y. (cf. the K-Ar age of 84.4). Magnetic ages at sites H and G must be very similar, and possibly slightly greater than the age at site F. The ages at sites A, B, C, and D are probably about the same, and they may all be greater than the age at site E. If site D has been appreciably reheated by the Hoffman monzonite, it could be younger than site A. The range given in Table 2 for Sentinel and Half Dome is probably maximum, in the sense that the actual variation within each pluton is less than the stated interval.

Absolute ages have an uncertainty of 1 to 2 per cent; they all fall in the early Upper Cretaceous. Our nine sites probably span about 10 m.y.; the average direction at each site probably represents the average field direction over periods of 10^5 years (see below).

Sampling and Laboratory Procedures

One hundred and seventy-nine oriented cores of granites and autoliths were collected at twenty sites in five of the map units, as shown in Figure 1. The cores were cut in the field with a portable gasoline-powered diamond drill; before the cores were broken loose their orientation was determined with a tube, slotted at its lower end and fitted with a hinged platform for a Brunton compass at its upper end. The cores are 0.980 inch in diameter. They were cut in the laboratory into one or more specimens 1 inch long.

Directions and intensities of magnetization were determined in the laboratory with a 'spin-

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TABLE 3. Average Values of Natural Remanence J_n, Susceptibility χ, and Q Factor for Nine Sites A to J, in emu
N = number of specimens.

Site	Granites				Autoliths				Average for All Specimens from Each Site		
	N	$J_n \times 10^4$	$\chi \times 10^4$	Q	N	$J_n \times 10^4$	$\chi \times 10^4$	Q	$J_n \times 10^4$	$\chi \times 10^4$	Q
A	2	3.8	38	.26	13	7.7	42	.40	7.2	41	.38
B	8	6.6	33	.44	13	9 3	32	.64	7.4	32	.62
C	10	4.1	29	.32	11	15	40	.89	9.8	34	.62
D	2	2.2	20	.25	4	4.1	40	.22	3.0	33	.22
E	4	7.5	43	.38	4	8.5	48	.40	8.2	45	.38
F	6	4.2	22	.43	3	3.8	17	.51	4.3	20	.46
G	2	.65	3.3	.44	24	.88	2.3	1.0	0.9	2.4	.95
H	3	1.2	9.3	.29	6	2.5	7.9	.78	2.0	8.3	.65
J	18	2.6	22	.26	2	4.5	28	.36	4.3	23	.26

ner' magnetometer of the type described by *Nagata* [1961, pp. 61–62, Figures 2–17 and 2–18], which had originally been constructed by *Doell* [1955].

Reversible weak-field susceptibilities were measured on a commercial alternating-current bridge. The electrically heated oven for the thermal demagnetization experiments was nonmagnetic and noninductive; the ambient field was maintained at less than 1 per cent of the earth's field, and a nitrogen atmosphere was maintained. Alternating-current demagnetization was done by tumbling the specimen about three axes simultaneously in a 60-cps alternating field which was continuously and smoothly reduced from its initial strength to zero over a period of 2 minutes by means of a motor-driven variable inductance.

Preliminary Evaluation of Data

Owing to a lack of internal consistency demonstrated by excessive scatter and/or by planar distributions of directions, data from a number of sites were discarded. In every case a site was either accepted or rejected in toto, even though this procedure resulted in discarding some apparently concordant data. Figure 2 gives two typical examples of sites yielding unacceptable data. Figure 2*a* is an example in which the scattered directions are unstable relative to the grouped directions during storage for 1 month in the earth's field in the laboratory. The site is an old road cut in El Capitan granite, on the Glacier Point road, and the origin of the scatter is not known. The scatter in Figure 2*b* is even greater and the distribution is crudely planar. The site is a glaciated surface on Half Dome quartz monzonite, and the directions appear to be related to the locations of the cores on the outcrop. The cores within a group are spaced a few inches apart, and groups are spaced about 20 feet apart on the outcrop. Many of the specimens from this site also had anomalously high intensities of remanence.

To facilitate comparison of intensities of magnetization among specimens containing varying quantities of ferromagnetic minerals, provided that the rock type, magnetic mineralogy, and thermal history are uniform for all, one can use Q, the ratio of the natural remanent intensity to the intensity of magnetization induced by the earth's field [*Koenigsberger*, 1938]. (Here $Q = J_n/H\chi$, where J_n is the intensity of natural remanence, H is the approximate intensity of the earth's field, taken as 0.45 in this latitude, and χ is the reversible volume susceptibility in a weak field.) In Figure 2*b* it is seen that Q is quite variable for this site and that it is anomalously high for many specimens, especially for those with directions in the upper hemisphere. The high Q values and the distribution over the outcrop of the scattered directions suggest remagnetization by lightning [*Cox*, 1961].

Another cause of scatter is slumping of outcrops, a condition which cannot always be detected in the field owing to lack of structure in the rocks. In particular, although the 'diorite and gabbro' have a strong and stable remanence, no accessible outcrops of this rock could be

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TABLE 4. (After *Roquet* [1954].) Susceptibility χ, Thermal Remanence, and Q Factor for Magnetite Dispersed in an Inert Matrix

	χ, emu	TRM in 0.42 oe	Q
Coarse-grained (average diam. of grain about 330 μ)	4.62×10^{-4}	0.95×10^{-4}	0.45
Fine-grained (diam. less than 5 μ)	1.27×10^{-4}	5.95×10^{-4}	10

found in place because of its rapid weathering relative to the granites.

In most cases, the sites rejected on account of excessive scatter showed either magnetic or geological properties that were significantly different from those of the accepted sites. For example, sites in the very fresh road cuts (1 or 2 years old) along the new Tioga road gave consistently good results, while some of the sites on glaciated surfaces adjacent to this road showed much scatter. Ultimately, data from 141 specimens cut from 76 cores obtained at 9 sites were retained, and data from 176 specimens cut from 103 cores obtained at 11 sites were discarded. The results from the Half Dome quartz monzonite, the Sentinel granodiorite, and the Gateway granodiorite were retained only in part, and all the results from the El Capitan granite and from the 'diorite and gabbro' had to be discarded.

Magnetic Properties

Table 3 shows the natural remanence J_n, the susceptibility χ, and the Q factor for each of the nine sites that were finally retained; mean values for granites and autoliths are given separately. These values fall within the usual range for granitic rocks and may be compared with *Roquet*'s [1954] values for artificial samples consisting of 1 per cent (by weight) of pure magnetite dispersed in an inert matrix (Table 4). The coarseness of the magnetite grains in our samples and the fact that they presumably have been well annealed during protracted cooling account for their low natural remanence and low Q factor. Differences between granites and autoliths are also explained by the different magnetite content and finer grain size of the latter.

Figures 3*a* and 3*b* illustrate the scatter of directions measured on specimens from two retained sites. Although the experimental error is probably not greater than 3°, directions of magnetization of duplicate specimens cut from the same core usually diverge by much more than this. Many specimens show pronounced instability; of 140 specimens that were measured shortly after collection in August 1961 and again in January 1962, 58 had changed by 5° or more, the average change of the latter being 11° and the maximum about 30°; the remaining 82 specimens, however, changed by an average of only 3°, which is close to the experimental error. Table 5 shows the mean declination and inclination at each site as first measured shortly after collection (columns *a*) and again after storage in random orientations in the earth's field for periods of 1 month (sites A, C, G) and 6 months (sites B, D, E, F, H, J) (columns *b*); it will be noticed that in spite of the large changes in direction that occurred in individual specimens, the average for each site changed only by very few degrees. Table 6 shows typical results for ac demagnetization of 13 randomly chosen specimens from site B. Table 7 shows similar results for thermal demagnetization of specimens from site G. It is noteworthy that, except for site B, demagnetization did not reduce the scatter; at eight of the nine sites, the highest value of the precision factor k corresponded to the measurements made either immediately after collection or after storage (see Table 5, column *c*).

Demagnetization could not, in general, be carried much beyond 100 oersteds or 200°C because of rapidly increasing errors in the determination of directions on individual specimens, as is indicated by excessively large 'error triangles.' These triangles [*Graham*, 1955] occur when three independent angular measurements of direction are plotted on a projection as three great circles which should intersect at a point representing the direction of magnetization. In practice, the three great circles form a triangle of closure or 'error triangle' which is usually such that the direction can be read within 1° or 2°. For some of our specimens, however, and particularly for those that had been demagnetized, the error triangles were extraordinarily large, as if the direction of magnetization in the specimen had changed while the specimen was being spun in the magnetometer.

We interpret the tendency for the precision

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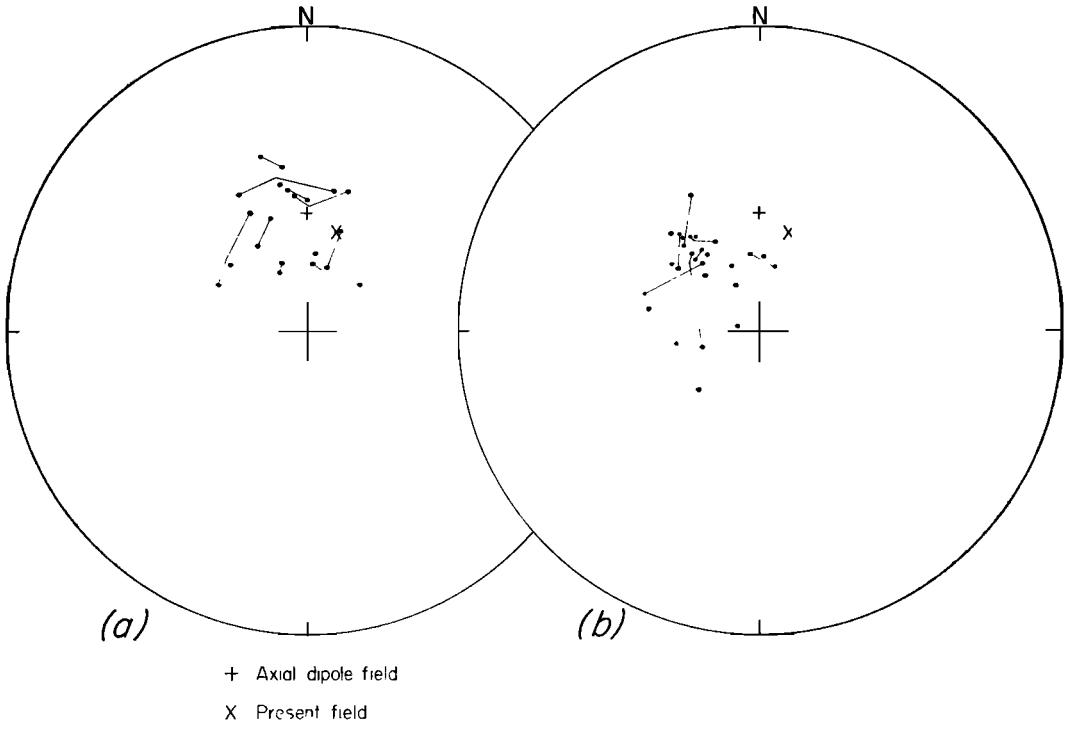


Fig. 3. Equal-area projections of north-seeking directions of magnetization of all specimens (*a*) from site C and (*b*) from site G. All points in lower hemisphere. Lines connect directions of specimens cut from one core.

factor k to decrease as demagnetization proceeds to mean that most specimens carry a 'random' component of magnetization (see below) which is insensitive either to alternating-current or to thermal demagnetization; as demagnetization gradually destroys the underlying TRM, the random component becomes relatively larger and the scatter increases (k decreases). Only for site B did demagnetization result in a substantial increase in k (Table 5, column *c*).

The demagnetization curves (Figures 4 and 6) are such that at first we doubted that our specimens carried any measurable thermal remanence at all. To test this possibility, nine specimens having an average natural remanence of 2.0×10^{-4} were heated to 620°C in nitrogen and cooled very slowly (2 days) in the earth's field. Their average remanence was then 3.3×10^{-4}. The same specimens were then reheated and given a partial TRM (PTRM) in the earth's field over the interval 620° to 450°C; the average PTRM was 2.1×10^{-4}.

Under ac demagnetization, the PTRM behaves much like the artificial total TRM and the natural remanence (see Figure 5) with regard to both the intensity and the directional stability. Artificial TRM decreases somewhat faster than the natural remanence.

Thermal demagnetization curves for natural remanence, artificial total TRM, and PTRM are given in Figure 6. As was mentioned, the natural remanence of autoliths appears somewhat more stable than that of granites. The initial (25° to 350°C) portion of the total TRM curve appears to be qualitatively similar to the curves for natural remanence. The latter were not continued above 350°C for the reason given above; thus they cannot be compared in the range 400° to 600°C with the curves for artificial TRM. From the curve for total TRM it is apparent that, for the autoliths at least, over half of the TRM is acquired between the Curie point (570°C) and 450°C. The PTRM curve serves merely to confirm the additivity law of thermoremanence for these rocks. In contrast to the natural remanence, directions of artificial TRM remain stable up to

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TABLE 5. Mean Directions at Each Site; *a* Immediately after Collection, *b* after Storage in Earth's Field for 1 or 6 Months, and *c* after Treatment Resulting in Highest *k* (see below)

Site	Declination, deg. west of north			Inclination, deg. down			N	k			α_{95}	
	a	*b*	*c*	*a*	*b*	*c*		*a*	*b*	*c*	*a*	*b*
A	39	43	*a*	67	65	*a*	15	62	56	*a*	5	5
B	16	11	16	63	66.5	62.5	21	27	29	49	6	6
C	9	12	*a*	59.5	60.5	*a*	21	32	28	*a*	5	6
D	17	32	*b*	46	47.5	*b*	6	28	50	*b*	11	8
E	5	5	*a*	61	61.5	*a*	8	44	43	*a*	7	7
F	3.5	9	*b*	56	54	*b*	9	27	38	*b*	9	8
G	42.5	46	*b*	66	67	*b*	26	35	36	*b*	5	5
H	58	59	*b*	59	57	*b*	9	56	66	*b*	6	6
J	6.5	1.5	*a*	63.5	62.5	*a*	20	17	16	*a*	8	8

500°C or higher. It is interesting that the thermal demagnetization curves for TRM and PTRM seem to extrapolate to a 'minimum' residual magnetization of about 0.4×10^{-4}.

Sources of Dispersion

The values of the precision factors for the nine sites listed in Table 5 fall in the range commonly observed in lavas and sediments. This is surprising, considering that most of our specimens probably remained in the temperature range in which most of the thermal remanence is acquired for periods of time of the order of 100,000 years or more. One would expect the secular variation to cancel out when averaged over such long periods, and the dispersion should be markedly less (i.e., k should be greater) for our granites than for rapidly cooled lavas or relatively rapidly accumulated sediments. It is likely that the within-site scatter is related to the unsteadiness, during demagnetization, of direction within single specimens. Possible causes of the observed scatter will be examined briefly.

Experimental errors. Directions measured on individual specimens are affected by an error of not more than 3°, as estimated from the triangles of closure on the stereographic projection [*Graham*, 1955]; errors in orientation and reference line inscription on the cores are less than 2° and in any case should not affect duplicate specimens from the same core. The divergence of directions on specimens from the same core is much greater than the experimental error.

Viscous magnetization. Viscous magnetization acquired in situ, before the specimens are

TABLE 6. Alternating-Current Demagnetization of 13 Specimens from Site B

	Declination, deg. west of north	Inclination, deg. down	k	α_{95}
Aug. 1961	9 5	66.5	27.3	7 4
Jan. 1962	18	59.5	29.3	7.2
Mar. 1962	20	60	23.1	8.1
After 10 oe*	19	60	31.6	6.9
20 oe	18	60	28.6	7.3
30 oe	18	60	26 7	7.5
40 oe	18.5	59	26.1	7.6
50 oe	17	59	23.5	8.0
60 oe	20	58 5	20 0	8.7
70 oe	16	57	19.7	8.7
80 oe	20	56.5	18.2	9.1
90 oe	18.5	58.5	13.9	11.3
100 oe	17.5	58	14.5	11.1

* Peak initial alternating field strength.

TABLE 7. Thermal Demagnetization of 26 Specimens from Site G

	Declination, deg. west of north	Inclination, deg. down	k
June 1962	42.5	66	35
July 1962	46.5	67	36
After heating*			
to 90°C	43 5	66.5	32
to 150°C	40.5	65 5	35
to 200°C	41	63	30

* In field-free space.

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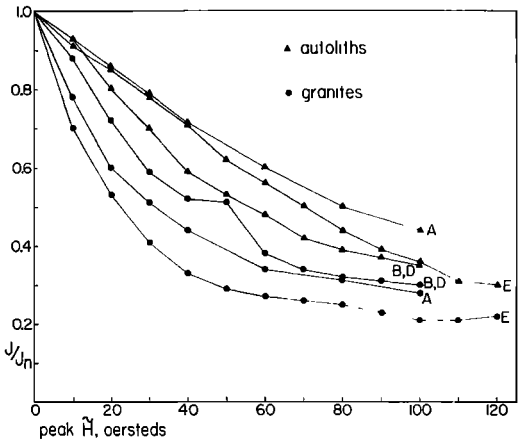


Fig. 4. Alternating-current demagnetization of natural remanence. Each curve is a normalized average of several specimens from the designated site. Results from granites and autoliths are plotted separately.

collected, should lead to an easily recognizable type of dispersion in which all directions lie near a plane containing the original direction of the remanence and the present direction of the earth's field. No such 'streaking' was observed. Viscous magnetization acquired during transportation and storage of specimens in the laboratory would tend to be random and would be relatively easily erased by alternating-current or thermal demagnetization. The fact, reported above, that the scatter increases as demagnetization proceeds would seem to exclude viscous magnetization as an important cause of scatter, although viscous magnetization is present in many of our samples. In fact, as is seen in Table 5, for two-thirds of the sites the higher k value is associated with a declination farther west of north, i.e., farther from the present field direction.

Random moments. *Irving et al.* [1961] have discussed the origin of the 'stray' magnetization often acquired by the specimens during magnetic cleaning: in high fields, the intensity does not diminish regularly with increasing field, but levels off to a constant value which they call the minimum intensity. They then consider a model in which magnetic domains behave as an assembly of antiparallel vectors having the values $+1$ and -1 with equal probability. There is a finite probability that the resultant of these vectors, which they call the 'random moment,' will not be zero. They deduce that the ratio M_r/M_s of the random moment to the saturation moment obtained when all vectors are parallel is

$$M_r/M_s = 0.76(n_1 n_2)^{-1/2}$$

where n_1 is the number of ferromagnetic grains per cm³ of rock and n_2 is the number of domains per grain. For a given volumetric fraction of ferromagnetic material, n_1 is proportional to a^{-3} (a is the grain size), while n_2 varies, for some simple configurations of the domains, as $a^{1/2}$ or a [*Craik and Tebble*, 1961]. Thus M_r/M_s may increase as a or $a^{5/4}$. On this model, the coarseness of the magnetite grains in the Sierra granites could account for a random moment greater than the natural remanence: a rough count in a granite showed, for instance, a number $n_1 = 10^8/\text{cm}^3$ of grains (a about 150 μ) in which the number n_2 of domains would be of the order of 30 and for which M_r/M_s is, accordingly, 4.4×10^{-3}. The saturation moment is about 1.3, as calculated from the volume proportion of grains (3.3×10^{-3}) and the saturation moment of magnetite (400); thus $M_r = 5 \times 10^{-3}$, which is clearly too large. The model proposed is unsatisfactory because it ignores the self-demagnetizing field that would substantially decrease the probability of nonzero values of M_r; yet it is clear

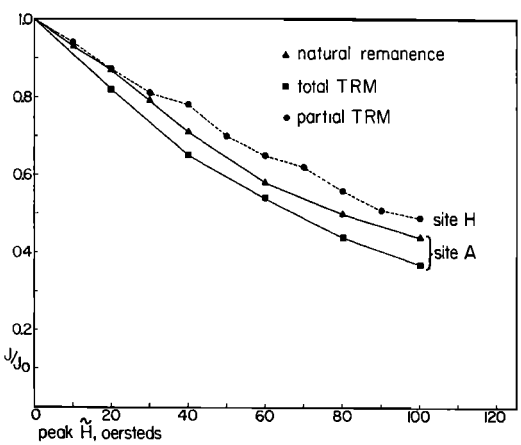


Fig. 5. Alternating-current demagnetization of natural remanence, artificial TRM induced by cooling from 620° to 25°C in the earth's field (0.45 oe), and partial TRM acquired in the same field over the temperature interval 620° to 450°C. Results from autoliths only. (PTRM curve is dashed because it is not strictly comparable to the other two, which were obtained on the same set of specimens.)

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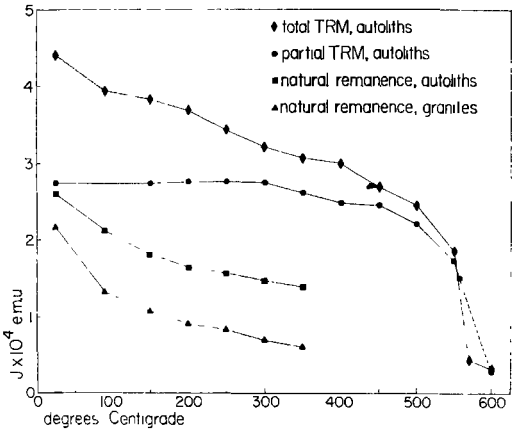


Fig. 6. Thermal demagnetization of natural remanence, artificial total TRM induced by cooling from 620° to 25°C in the earth's field (0.45 oe), and partial TRM acquired in the same field over the temperature interval 620° to 450°C. (All measurements made at 25°C.)

that in coarse-grained rocks an appreciable random magnetization could exist, caused by an imperfect balance of variously oriented domains.

The fact noted above (see Table 6) that the mean direction of a group of specimens hardly changes as demagnetization proceeds, while the scatter markedly increases, strongly suggests the presence of a random component of magnetization of nearly constant intensity superposed on a directed component, the intensity of which decreases as demagnetization progresses. We have followed in some detail the ac demagnetization of two specimens from site B. First the average decrease in intensity as a function of the ac field was determined for all specimens from the same site. After each step in the demagnetization process both the intensity and the direction were noted, and successive steps were compared. We then assumed that changes from one step to the next were due (1) to a decrease in intensity of the directed component equal to the average decrease for the group for that step and (2) to the addition of a component for which we calculated the intensity and angle α (Figure 7). For one specimen the magnitude of eight of these vectors, corresponding to eight successive steps of 10 oersteds each, ranged from 0.1 to 0.4 $\times$ 10^{-4}, with an average of 0.23 $\times$ 10^{-4}, and the angle α ranged from 2° to 140°, with an arithmetic mean of 88°. For another specimen, the average (for ten steps) of the angle α was 91.5°, and the average intensity was 0.19 $\times$ 10^{-4}. The natural remanence of these two specimens was 3.79 and 3.81 $\times$ 10^{-4}, respectively. We interpret these results to mean that these specimens carry a randomly oriented magnetization of nearly constant intensity (0.2 $\times$ 10^{-4}).

A. Cox (personal communication) has calculated the precision factor k for a population of vectors, each of which is the sum of a fixed component of unit intensity with precision factor $k_1 = \infty$ and a random vector of magnitude x; he finds that $k \approx 3/x^2$. For the specimens described above $x = 0.2/3.8 = 0.052$, and thus $k = 1000$, which implies a dispersion much less than the measured dispersion at the site and suggests that the random magnetization does not account for all of the dispersion. This can also be seen by comparing values of k at different stages in the demagnetization process. Consider a typical site at which the initial precision factor k_0, before demagnetization, is 30. If the scatter is to be entirely explained by a random component R, we find from Cox's relation that $R = 0.32J_0$, where

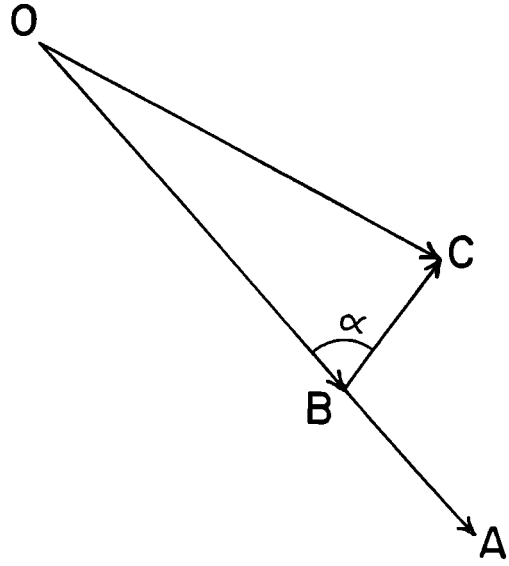


Fig. 7. Form of vector diagram used to calculate random magnetization acquired during ac demagnetization. OA is the magnetization of one specimen after the nth step in demagnetization, OC is the magnetization after the $(n + 1)$th step. The length AB is calculated from the average percentage decrease in intensity for the entire group from the nth to the $(n + 1)$th step. Thus BC is the added component.

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TABLE 8. Mean Pole Positions and Statistical Parameters

	Average Coordinates of Sampling Sites	Declination, deg. west of north	Inclination, deg. down	N	B	ω	β	K	α_{95}	Pole Positions Lat.	Pole Positions Long.	δ_p	δ_m
a	37°46′N 119°34′W	N 21.5 W	+62.5	134	9	34	179	1188	4.1	73.5°N	174°E	7.3	9.5
b		N 25 W	+61	133	9	30	202	1250	4.0	70.5°N	171°E	7.5	9.8

J_0 is intensity of the fixed component and $x = R/J_0$. When the demagnetizing field reaches 100 oersteds, the precision factor k_{100} may be as low as 15; hence $R = 0.45J_{100}$. If R is independent of the demagnetizing field, $J_{100}/J_0 = 0.32/0.45 = 0.7$; actually we measure values of J_{100}/J_0 between 0.3 and 0.5. An artificial ac demagnetization curve constructed in this manner from the variation of k drops off less rapidly than the observed curve. We conclude that the random moment accounts for part, but not all, of the observed scatter.

Secular variation. Observed rates of secular variation suggest that the geomagnetic field, when averaged over a period of the order of 10^4 years, would correspond very closely to that of a dipole aligned along the rotation axis [*Creer*, 1962]. *Doell and Cox* [1961*a*] note that paleomagnetic results made from widely separated areas indicate that the geomagnetic field during the past half-million years, when averaged over an interval of time of the order of 10^4 to 10^5 years, has very closely approximated the field of a geocentric dipole. The mean position of the pole for the past half-million years is indeed extraordinarily close to the present rotational pole.

As was mentioned above, most of our specimens have probably spent several hundred thousand years at temperatures between 600° and 400°C. All specimens from a site come from within a few inches or, at most, a few feet of each other. As a typical rate of displacement of isotherms may be about 0.4 cm/year, the magnetic ages of the several specimens from a single site cannot differ by much more than 10^3 years; specimens from the same core could not differ in age by much more than a few years. One would not expect the direction of the field when averaged over an interval of 10^5 years to differ significantly from that for an equal interval starting, say, 10 years later. Secular variation thus seems an unlikely source of scatter.

It should be kept in mind, however, that TRM is the vector sum of a number of partial remanences acquired in successive temperature intervals. Partial remanences are quite small at low temperature and also very close to the Curie point; in between there is a peak which, if sharp, means that a large fraction of total TRM is acquired in a small temperature interval and therefore in a relatively short time. If this is the case for the Sierra Nevada granites, many specimens may have acquired most of their TRM in 10^4 rather than 10^5 years. Nevertheless, if the secular variation has contributed to the dispersion, it may be necessary to conclude that the average direction of the field (for intervals of 10^4 years) has undergone sudden changes of the order of a few degrees. Recent data from Hawaii suggest that the field did behave in this manner during the Pleistocene [*Doell and Cox*, 1961*b*].

Inhomogeneity and anisotropy. It remains to consider whether the observed scatter could be due to the inhomogeneity of the granites. Scatter could, of course, occur if the rocks were anisotropic and if, in addition, directions of anisotropy changed markedly within a few inches. There is, however, no visual evidence of anisotropy. Under the microscope most magnetite grains appear to be equidimensional. Occasionally one does see rows of small grains strung along the c axis of hornblende crystals, but the hornblende crystals appear to be distributed isotropically. Autoliths may tend to be ellipsoidal, but the resulting anisotropy cannot have any serious effect on the direction of remanence, as the intensity of magnetization is very weak: the maximum demagnetizing field ($4\pi J_n$) corresponding to the most adverse conditions would still be only a small

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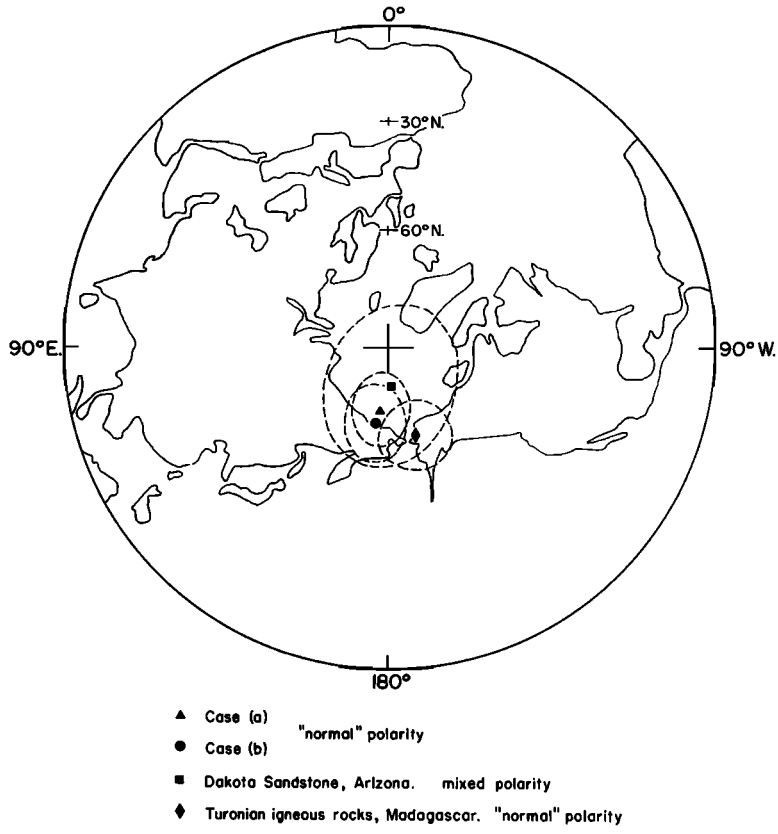


Fig. 8. Polar equal-area projection on northern hemisphere of mean pole positions and ovals of confidence from Table 8, cases (a) and (b), together with pole positions from the Dakota sandstone and from Turonian igneous rocks in Madagascar.

fraction of the earth's field. As was mentioned earlier, sites were selected with care to avoid outcrops which showed distinct foliation or lineaton. The direction of artficial TRM was, whenever checked, invariably found to be parallel to the external field, within the experimental error. It thus appears unlikely that inhomogeneity or anisotropy could contribute to the scatter.

To summarize, the within-site scatter exceeds the experimental error. It may be caused in small part by a viscous magnetization acquired between collection and first measurement of the specimens. Most specimens seem to carry a 'random magnetization,' due perhaps to imperfect balance of domains, which accounts for part of the scatter. It is also possible that the average (over 10^4 years) field direction at each site underwent sudden changes of the order of a few degrees.

Pole Positions

The mean pole position and statistical parameters for the nine sites have been calculated using the two-level sampling scheme of *Watson and Irving* [1957] and are given in Table 8. Here $\alpha_{95} = 140/\sqrt{K}$ and $K = [(\omega N)^{-1} + (\beta B)^{-1}]^{-1}$, where N is the total number of specimens collected at B sites, ω is the within-site precision parameter at all sites, and β is the between-site precision parameter. Two sets of results are given. The first (a) is based on the first measurements made immediately after collection (Table 5, case *a*). The second set (b) corresponds to case (*c*) and Table 5; i.e., it was calculated by selecting for each site the data corresponding to whatever stage in the treatment of specimens gave the highest k for the site (cf. Tables 6 and 7). Values of K and α_{95} in Table 8 refer to the mean direction for all specimens,

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TABLE 9. Statistics for the Mean Pole Positions Given in Table 8, Calculated by Treating Each Site-Mean as a Single Observation

	N	R	k	α_{95}	$\delta = \cos^{-1} R/N$, deg.
a	9	8.857	56	6.2°	10.2
b	9	8.852	54	6.4°	10.4

whereas the directions listed in the table are means of the nine site-mean directions. Over-all mean and mean of site-means would be identical if the number of specimens were the same at all sites. The difference is not important.

Figure 8 shows the mean pole positions and ovals of confidence from Table 8, cases (a) and (b), together with the only published Cretaceous pole for North America (Dakota sandstone [*Runcorn*, 1956], quoted by *Cox and Doell* [1960]) and a pole obtained from Turonian (lower Upper Cretaceous) igneous rocks in Madagascar (quoted by *Cox and Doell* [1960]).

Statistics corresponding to the case in which the mean directions at each site are treated as single observations are given in Table 9, in which we have included the angular dispersion factor δ for comparison with some of *Creer's* [1962] results (see below).

Figure 9 shows (*a*) the site-mean directions and the mean of these with the associated circles of confidence and (*b*) the virtual geomagnetic poles corresponding to these, all taken from Table 5, case (*c*), and Table 8, case (b).

If the secular variation, including that of the equatorial component of the dipole, is effectively smoothed out by the length of time during which these specimens remained at high temperature, the angular dispersion factor δ must refer to the motion of the rotational pole. Our sites cover a total span of about 10^7 years, and some of them may overlap (e.g., A, B, C, D). Figure 9*b* then implies quite large and sudden shifts of the pole, of the order of 10° per million years. The average rate of wandering of the rotational pole inferred from British and North American rocks is about 0.6° per million years between the late Precambrian and the present time [*Creer*, 1962]; the present rate of polar displacement, if real, is less than about 10 feet toward Greenland in the last 50 years, or 0.5° per million years. The rate implied by our data thus seems unreal, and it is preferable to suppose that the virtual poles for each site are paleomagnetic, not paleogeographic poles.

Two curious features appear in Figure 9*b*. In the first place, the virtual poles are not randomly distributed about the mean pole; instead they lie approximately on an ellipse whose major axis is about 30° long and is parallel to the meridian of the mean pole. Its minor axis is about 20° long. In the second place, if we suppose that the true chronological order of magnetization of the sites is J-G-H-F-E-D-C-B-A from oldest to youngest, the virtual poles fall in chronological sequence in 2¼ revolutions *clockwise* around the mean pole. This sequence of magnetization corresponds very closely to that previously inferred from geologic evidence alone, which may not be accidental. (For instance, consider the virtual poles A, B, C, and D. If we assume that the elliptical path is real, the chance of these poles falling in any single given order on it is only 1 in 24.) This in turn could mean that the geomagnetic pole moved clockwise (i.e. from east to west) around the geographic pole, a movement which would correspond to a westward drift of the equatorial component of the dipole moment, the relative intensities of the polar and equatorial components remaining nearly constant. The period of revolution would be approximately twice the age difference of sites D and A, or a little less than twice the age difference of sites E and G. It should be remembered that the estimates of the times spanned by the groups of sites in each pluton are based entirely on calculated rates of cooling which cannot be evaluated precisely because of our ignorance about physical conditions prevailing at the time of intrusion. It is unlikely that the cooling rates would be slower than these estimates, but they might be several times faster. Thus the period of revolution should not be greater than 2×10^6 years, but it is probably not less than 10^5 years. If this interesting result is supported by further studies of these and similar plutons, it will be necessary to conclude that a virtual magnetic pole corresponding to the mean direction of magnetization of a formation is identical to the paleogeographic pole if and only if the span of time represented by the remanent magnetization of the formation exceeds 10^6 or 10^5 years, not 10^4 years as *Creer* [1962] concluded.

It is noteworthy that the angular dispersion

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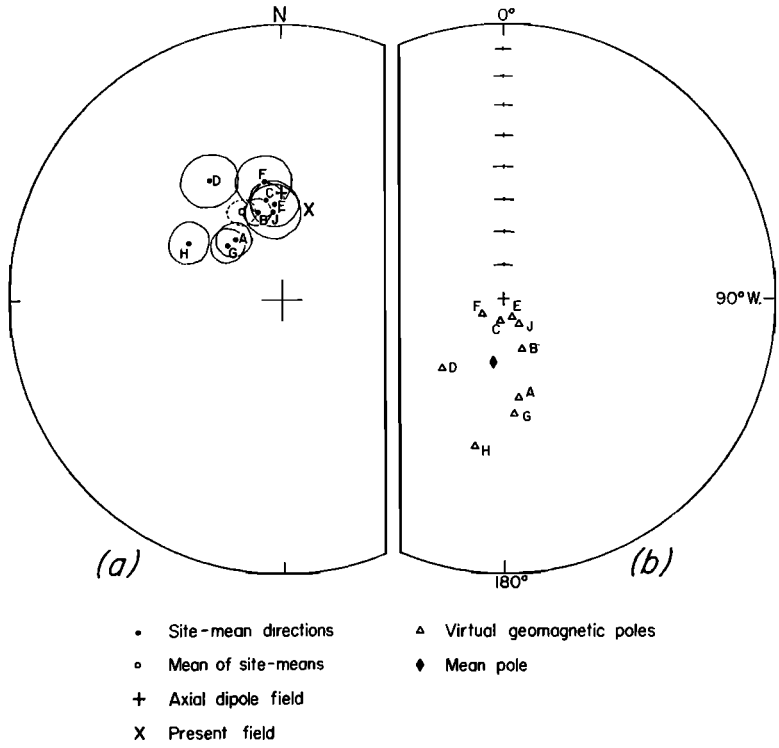


Fig. 9. (*a*) Equal-area projection on lower hemisphere of mean north-seeking directions of magnetization at each site and over-all mean direction, with circles of 95 per cent confidence. (*b*) Polar projection on northern hemisphere of corresponding virtual geomagnetic south poles. Data taken from Table 5, case (*c*), and Table 8, case(b).

factor δ for our poles is only slightly less than that (about 12°) which would be appropriate to the mean paleomagnetic latitude of our sites, as deduced from paleomagnetic data [*Creer*, 1962, Figure 10] for flows, dikes, and sills, all of which have had a much shorter cooling history than the Sierra Nevada plutons. It follows either that the Sierra Nevada plutons have acquired their magnetization in a time much shorter than 10^5 years or that the time constant for the secular variation of the dipole field is closer to 10^6 than to 10^4 years.

As more K-Ar ages are determined, it may be possible to reconstruct more exactly the cooling history of some of the plutons. We propose to continue paleomagnetic work on these.

Summary and Conclusions

A large number (about 300) of specimens from twenty sites distributed in five granitic plutons of early Upper Cretaceous age in Yosemite National Park have been examined. Eleven sites were discarded on the basis of evidence of slumping, or magnetization by lightning, or because of excessive scatter; nine sites (134 specimens) were finally retained.

The magnetic constituent of all rocks is, uniformly, a very pure magnetite. The Q factors are low, probably because of the large size and annealed state of the magnetite grains. About one-third of the specimens showed instability; directions of some specimens changed appreciably (up to 20° or more) by storage in the earth's field, or during either alternating-current or thermal demagnetization. These changes seem, however, to be quite random; although the scatter at any one site increases notably as demagnetization proceeds, the average direction for that site hardly changes at all. Two mean poles have been computed. The first is based on data measured soon after collection; the second is based on the data for each site corresponding to that particular step of the magnetic cleaning which gave the least dispersion for that site. The two poles are not significantly different; both differ significantly from the present geomagnetic

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pole. The source of the within-site scatter was examined; it seems to be due partly to unexplained random magnetic vectors which are probably not purely of viscous origin.

Virtual poles were computed for each site. They are distributed with an angular dispersion factor $\delta = 10.2°$, which is about the value (for the appropriate paleomagnetic latitude) that *Creer* [1962] finds for post-Carboniferous lava flows, dikes, and sills. This result is puzzling. Each of our specimens may have remained in the temperature range from the Curie point to 400° for times of the order of one to several hundred thousand years, and the secular variation should accordingly be effectively smoothed out. Our results seem to imply that the geomagnetic pole for a formation can be identified with a paleogeographic pole only if the sampling covers a span much greater than 10^5 years. The mean pole calculated from our measurements, which cover 10^7 years, may be a paleogeographic pole. Individual site poles, which cover perhaps as much as 5×10^5 years, are not believed to be paleogeographic poles. If they are paleogeomagnetic poles, it may be inferred from Figure 9*b* that the geomagnetic pole moves from east to west around the geographic pole with a period of the order of 10^5 to 10^6 years.

Our conclusions should, however, be tempered by the observations (1) that the within-site scatter of directions and the instability of some of the specimens remain unexplained; (2) that structural conditions at the time of intrusion of these plutons are not well known; no reliable estimate has yet been made, for instance, of the thickness of the sedimentary cover, with the result that the cooling history is only imperfectly known; and (3) that the chronological order of some of the sites is uncertain. It is hoped that further geological and paleomagnetic work on these Sierra Nevada plutons may provide more definite results.

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